INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

THREAD, POLYESTER

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for all federal agencies.

1. SCOPE. This Commercial Item Description covers the requirements for polyester thread used for machine sewing.

2. CLASSIFICATION. The following types, classes and subclasses of threads shall be as specified.

2.1 Types.

Type I - Twisted, soft, multiple cord
Type II - Twisted, bonded, multiple cord
Type III - Bonded, monocord
Type IV - Soft, air entangled
Type V - Bonded, air entangled

2.2 Classes.

Class 1 - Normal elongation.
Class 2 - DELETED
Class 3 - Heat stable – low shrinkage

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AMSC N/A FSC 8310

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2.3 Subclasses.

Subclass A - General purpose
Subclass B - Non-wicking finish
Subclass C - No finish

2.3.1 Subclasses references. In end item documents where no Subclass designation “A”, “B”, or “C’ is indicated, the requirements for Subclass “A” shall apply.

3. SALIENT CHARACTERISTICS.

3.1 Description. All threads shall be bright, continuous multifilament polyester yarn with a density of 1.36 to 1.40 grams/cubic centimeter (g/cc). The bonding agent used in Types II, III, and V shall be colorless. Unless otherwise specified, the direction of the final twist shall be “Z” for Types III, IV and V. For Types I and II thread (used for sole stitching of footwear) shall be “S” or “Z” depending upon the initial twist. The final twist for Type I and II shall be the opposite of the initial twist.

3.1.1 Color. The color shall be as specified in the applicable end item specification or in the contract (see 7.8). The finished thread shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 7.4).

3.1.1.1 Visual shade matching. The color and appearance of the thread shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 7.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the thread shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 7.5.1). The thread shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

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3.2 Physical requirements. The finished thread shall conform to the applicable physical characteristics as specified in Tables I, II, and III for types and classes, when tested as specified in Table IV.

TABLE I. Physical characteristics - Types I, II, and III, for Class 1.

Tex (sizes)	Government Letter Size or Number Size	Breaking strength, lbs. (min.)			Number of Ply		Elongation % (max.)
		Type I	Type II	Type III	Type I & Type II	Type III	Type I, II, and III
16	00	1.4	1.5	1.4	2	1	28
21-24	A	2.0	2.3	2.6	2 or 3	1	28
3035	AA	3.0	3.0	3.0	2, 3, or 4	1	28
45-50	B	4.3	4.5	5.3	2 or 3	1	28
70	E	8.0	8.0	8.0	3	1	28
90	F	10.6	10.6	10.6	3 or 4	1	28
135-150	FF	16.0	16.0	16.0	3 or 5	1	28
210	3	24.0	24.0	24.0	3	1	28
270-300	4	32.0	34.0	32.0	3	1	28
350	5	40.0	42.0	40.0	3	1	28
400-450	6	48.0	48.0	48.0	3	1	28
500	7	54.0	54.0	54.0	3	1	28
600	8	60.0	60.0	60.0	3 or 5	1	28
700	10	70.0	70.0	70.0	3, 5, or 6	1	28

TABLE II. Physical characteristics - Type I, Class 3.

Tex (sizes)	Government Letter size or Number size	Number of Plies	Breaking Strength, lbs.(min.)	Elongation %, (max.)
16	00	2	1.2	35
24	A	2 or 3	1.9	35
50	B	2 or 3	3.8	35
70	E	3	6.8	35
90	F	4	7.6	35
135-150	FF	3	16.0	35
210	3	3	19.0	35
300	4	3	27.2	35
350	5	3	34.0	35
450	6	3	41.0	35
500	7	3	47.7	35

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TABLE III. Physical characteristics - Types IV and V, Class 1.

Tex (sizes)	Breaking strength, lbs. (min.)	Number of Plies	Elongation %, (max.)
	Type IV and Type V	Type IV and V	Type I, II, and III
16	1.0	1	26
21	1.4	1	26
24	1.6	1	26
27	1.8	1	26
30	2.0	1	26
40	2.8	1	26
50	3.6	1	26
60	4.3	1	26
80	5.4	1	26
90	5.9	1	26
105	6.9	1	26

TABLE IV. Physical requirements and end item tests – (all types and classes).

Characteristic	Requirements	Test Method
Fiber identification	3.1	AATCC TM20
Density	3.1	AATCC TM20
Bright continuous multifilament	3.1	Visual
Visual shade matching	3.1.1.1	Table IV
Tex (size)	Tables I, II and III	ASTM D204 1/
Ply	Tables I, II and III	Visual
Colorfastness (all types and classes): Laundering (after 3 cycles) Perspiration Light (after 40 hrs. or 170kJ/) Dry heat at 356 (±) 3°F Weathering, Types I and III only, (after 80 hrs.)	 3-4 (min.) 3-4 (min.) 3-4 (min.) 3-4 (min.) 3-4 (min.)	 AATCC TM61 3A 2/, 3/, 4/ AATCC TM15 2/, AATCC TM16.3 2/ AATCC TM117 2/, 3/ AATCC TM169 Opt. 1 2/

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TABLE IV. Physical requirements and end item tests – (all types and classes).- Continued

Characteristic	Requirements	Test Method
Final twist	3.1	ASTM D1423
Breaking strength	Tables I, II, and III	ASTM D204 5/
Elongation	Tables I, II, and III	ASTM D204
Heat shrinkage (Class 3 only), %, (max.)	3.3	ASTM D204 6/
Resistance to wicking (Subclass B only)	3.4.1.1	3.4.1.2

1/ Tex sizes for all threads based on weight in grams/1,000 meters thread, per ASTM D204.
2/ AATCC EP1, Gray Scale for Color Change.
3/ The color transfer cloth evaluation shall not apply.
4/ The specimen shall be dried after each laundering cycle.
5/ Except that five (5) determinations shall be made per sample unit.
6/ Five (5) specimens prepared as specified in ASTM D204 shall be tested in accordance with the dry heat method and exposed for 1 hour using the weights specified in Table IV. The shrinkage shall be determined as a percent change from original measured length and the average test result of the five (5) specimens reported to the nearest 0.1 percent.

3.3 Heat stability shrinkage (Class 3 only). The Class 3 thread shall have a maximum shrinkage of 3.0 percent, when tested in accordance with Table IV and the weights in Table V.

TABLE V. Heat stability shrinkage test weights - Type I, Class 3 only.

Tex size	Weight, ounces (oz.)
16, 24, 50	1
70, 90,135-150	2
210	4
300, 350,450,500	8

3.3.1 Knots. Thread Tex size 150 and finer shall average not more than one (1) thread knot per 2-ounces, and thread Tex size 210 and heavier shall average not more than one (1) thread knot per 4-ounces.

3.4 Finishing materials. The finished thread shall have no chemical finishes or treatments other than those commonly used (e.g. water-repellent, etc.) on commercial threads or as specified in the contract which have been demonstrated to have no harmful effects on the fiber, including effects of prolonged storage. No finish or treatment shall be applied for the purpose of increasing breaking strength. There shall be no noticeable wicking of the treatment on the thread to adjacent material when sewn.

3.4.1 Bonding agent. Bonding agents shall be applied to Types II, III, and V, and when necessary to provide good sewing quality.

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3.4.1.1 Non-wicking finish (Subclass B). Subclass B thread shall have a uniformly applied commercial nontoxic non-wicking finish. The use of compounds containing mercury in any form shall not be used. The finished thread shall resist the wicking of water when tested as specified in 3.4.1.1 and 3.4.1.2.

3.4.1.2 Initial vertical resistance to wicking. The thread shall be water repellent treated so that the treated thread shall resist the wicking of water for a period of not less than 2-hours when tested as follows: The test specimen shall consist of a 20 strand skein of thread in one continuous 30 yard length made on a 54-inch periphery skein reel. The skein shall be reeled under enough tension to cause the strands in the skein to lie uniformly, side by side, on the reel. The finishing end of the skein shall be tied to the starting end of the skein in such a manner that the knot will not add additional length to the reel skein. The skein shall be hung over the movable crossbar of a laboratory stand with the end hanging over the vessel. The movable crossbar shall rise 28-inches or more above the base. A non-ferrous 3/4 to 7/8 ounce weight shall be placed in the lower catenary of the skein to keep it taut and straight. The skein shall be arranged so that the strands are touching each other in flat ribbon form. The vessel shall be filled to a depth of at least 5inches with distilled water at a room temperature which has been mixed with 0.05 percent blue food coloring (salt and wetting agent free). A piece of blotting paper shall be attached by means of a paper clip or similar clamp to one (1) full side (20 strands) of the skein, 3-inches above the lower catenary of the skein. The position of the crossbar shall be adjusted that when the skein is hung freely in the liquid, 2-inches of the skein will be immersed in the liquid and the lower edge of the blotter is 1-inch above the liquid surface. The skein shall then be slowly lowered into the dyebath and the time of entry shall be noted. Depending on the dimensions of the vessel and the length of the crossbar, several specimens can be tested at the same time in the same dyebath by hanging the skeins sufficiently apart on the crossbar. The skein shall be exposed for 2-hours. The blotter shall be examined for wetting or staining at least every hour. The test shall be terminated whenever staining or wetting of the blotter is observed within the 2-hour duration. Staining or wetting before the 2-hour time frame shall constitute a failure.

3.5 Put-up. Unless otherwise specified, the thread shall be put-up on holders such as commercial spools, cones, or tubes as specified in the contract. The thread shall be wound around the specified holder in one continuous piece, so that each turn and layer is free of entanglement. The outside ending of the thread shall be secured to prevent unwinding, loosening, or slippage during handling, shipping, or storage.

3.6 Labeling. Each thread holder shall have a label, adhered securely to remain in place and be clearly legible until all thread has been removed. The label shall be printed and include information related to length in yards or weight of cone, direction of twist, color, Tex size, name of thread manufacturer, and nomenclature specifying fiber type and construction. Type and class, stock number, date of manufacture (month and year) shall be on the label.

3.7 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished thread shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.7.1.

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3.7.1 Toxicity test. When the toxicity requirement (see 3.7) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 7.1.4 and 7.8).

3.8 Workmanship. The finished thread shall conform to the quality of the product established in this document. The thread shall average not more than one (1) full thread knot or splice per 1,000 yards. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data in the procurement documents.

4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 13834; as applicable. (See 7.1.2).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The thread polyester shall meet the salient characteristics of this Commercial Item Description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The finished thread polyester offered must have been sold to the government or commercial market.

5.3 End item inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.2).

5.4 Visual examination. Each finished thread polyester shall be examined for the defects listed in Table VII as defined in ASTM D3990. The lot size shall be expressed in units of thread polyester. The sample unit shall be one (1) thread polyester.

TABLE VII. Visual examination defects.

Knots: Size 150 and finer thread not more than one (1) thread knot per 2-ounces. Size 3 and heavier not more one (1) thread knot per 4 ounces of thread.
Color: Not as specified.
Workmanship: More than one (1) full thread knot or splice per 1,000 yards; defects exceed the quality assurance standards (both contractor’s and government standards).
Labels: Label missing, incorrect, or illegible. Required information missing from the label.
Packaging: Not packaged in accordance with the contract or purchase order

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5.5 Acceptance criteria. Acceptance criteria shall be as specified in the contract or procurement document (see 7.8).

6. PACKAGING. The preservation, packing and marking shall be as specified in the contract or order (see 7.8).

7. NOTES

7.1 Source of Government documents:

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.1.2 EPA and EO 13834 information is available online at https://www.epa.gov and https://www.fedcenter.gov/programs/eo13693.

7.1.3 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov.

7.1.4 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1).

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100).
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

7.2 Source of non-Government documents.

7.2.1 AATCC Test Methods are available for purchase online at https://aatcc.org.

7.2.2 ASQ/ANSI Z1.4 – Sampling Procedures are available online at https://asq.org.

7.2.3 ASTM Standards are available for purchase online at https://www.astm.org.

7.3 Intended use. The thread is intended for hand and machine sewing of clothing, equipage, and footwear. The Type I, Class 3 (Subclass A) thread is specifically intended for use in parachutes; that are subject to exposure at elevated temperatures. Subclass B threads are intended for tentage, equipage, and other items where water resistance of seams is required and used primarily in Type I and Type III threads. Type I and Type II threads are used for stitching of footwear. Subclass C thread covers the larger size of thread used in footwear and other leather items which are waxed at the sewing machine and require that no sewing or other type finish be applied.

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7.4 Standard samples. For access to standard shade samples of thread, address the contracting activity issuing the invitation for bids or request for proposal.

7.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

7.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222's should reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

7.6 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for a stated requirement.

7.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issues due to the extensiveness of the changes.

7.8 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this Commercial Item Description (CID)
b. Type, and class required (see 2 thru 2.3.)
c. Color required (see 3.1.1)
d. Physical requirements (see 3.2)
e. Put-up required if other than specified (see 3.5)
f. When toxicity testing is required (see 3.7and 3.7.1)
g. Product conformance provisions (see 5.1)
h. Acceptance criteria (see 5.5)
i. Packaging requirement (see 6)

7.9 Key words.

Cover, helmet
Equipage
Flag
Footwear
Parachute
Parka
Shoe
Streamers
Trousers

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MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Custodian:
Army - GL
Navy - NU
Air Force - 11

Preparing Activity
DLA –CT

(Project: 8310-2025-005)

Review Activities:
Army - MD

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